

Selective Control of Ion Transport by Nanoconfinement: Ionic Liquid in Mesoporous Resorcinol–Formaldehyde Monolith

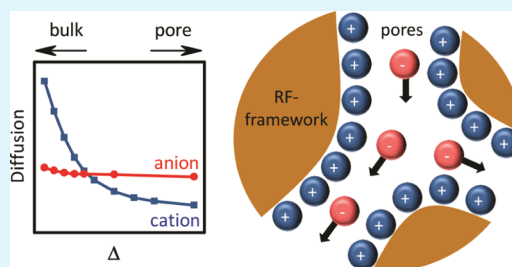
Carl-Philipp Elverfeldt, Young Joo Lee,* and Michael Fröba*

Institute of Inorganic and Applied Chemistry, University of Hamburg, Martin-Luther-King-Platz 6, 20146 Hamburg, Germany

Supporting Information

ABSTRACT: Thermal and dynamic properties of ionic liquid (IL)-based electrolytic solution (Li^+TFSI^- in $\text{Pyr13}^+\text{TFSI}^-$; 1-methyl-1-propylpyrrolidinium bis(trifluoromethylsulfonyl)imide = $\text{Pyr13}^+\text{TFSI}^-$) confined in nanoporous polymer hosts were investigated with respect to the pore size/porosity and the surface chemistry of the polymer host. As host material, mesoporous resorcinol–formaldehyde (RF) polymer monoliths with three-dimensionally connected pore structure were prepared, with precise control of the pore size ranging from ca. 7 to 60 nm. Thermal analysis of RF polymer–ionic liquid composites showed stability up to almost 400 °C and a melting point depression proportional to the inverse of the pore diameter. Good ionic conductivity comparable to that of a commercial separator is obtained, which is dependent on the porosity (i.e., pore volume) of the confining host material (i.e., the number of charge carriers available in the system). Further pulsed field gradient (PFG) NMR experiments revealed that the diffusion coefficient of Pyr13^+ cation becomes smaller than that of TFSI^- anion inside RF pores, which is contradictory to the bulk IL system. This change in the ionic motion is due to electrostatic attraction between the pore walls and Pyr13^+ cations, resulting in a layer structure composed of a Pyr13^+ cation-rich layer adsorbed at the pore wall surface and a TFSI^- anion-enriched bulklike layer at the pore center. Our study suggests that transport characteristics of the ions of interest can be controlled by optimizing the surface chemistry of the host framework and their motion can be separately monitored by PFG NMR spectroscopy.

KEYWORDS: Resorcinol-formaldehyde, ionic liquid, conductivity, diffusion, PFG NMR, confinement effect, mesoporous polymer



1. INTRODUCTION

Porous resorcinol–formaldehyde (RF) polymers have been known for over two decades and are still receiving substantial attention in the literature.^{1–6} The synthesis, similar to the sol-gel synthesis of inorganic oxides, consists of a hydrolysis-condensation mechanism: resorcinol (R) and formaldehyde (F) are mixed with a catalyst (C, can be basic or acidic) in a mainly aqueous solvent, followed by heating the solution in a closed container to form a cross-linked gel via a phase separation mechanism. Hence, the resulting gel is influenced by multiple parameters. These can be summarized as five main variables: (i) the dilution ratio of R and F to the total solution;⁷ (ii) the R/F ratio;⁸ (iii) the catalyst concentration,⁷ i.e., the pH value of the solution; (iv) the drying conditions, as they affect the shrinkage of the resulting porous polymer, and (v) the amount of organic solvent added and its polarity^{9–13} (methanol is used as a stabilizer in formaldehyde solution and thus always present in small quantities). By varying these parameters, it is possible to adjust the porosities and surface areas of the resulting polymer monoliths as well as other characteristics, i.e., thermal conductivity¹⁴ or flexibility.¹⁵ Besides the high porosities and surface areas of these porous RF gels, their cheap, easy, and fast synthesis route is one major advantage. Hard- or soft-templating synthesis routes on the contrary require expensive surfactants or a cumbersome synthesis of the hard template prior to the polymerization

step. Afterward, the template needs to be removed, which is often carried out with hazardous chemicals, such as concentrated sodium hydroxide or hydrofluoric acid (for the removal of a silica host). Also, due to these sacrificial templates, an upscale is often difficult. In contrast, the residual solvent can be easily removed from the RF gels by subcritical-, supercritical- or freeze-drying, resulting in a xerogel, aerogel, or cryogel, respectively, or by a simple solvent exchange. In this work, a hydrochloric acid-catalyzed route^{9,13} was employed in which the pore size generated is controlled by the amount of ethanol added to the starting solution. Usually the pH⁷ and the dilution of the reactants¹⁶ are the main factors used to control the final porosity, but the addition of an organic solvent (e.g., methanol^{10,11} or ethanol^{9,12,13}) surmounts these factors and makes it even simpler to adjust and predict the resulting pore size. Combined with subcritical drying, a washing step for the removal of the catalyst or a solvent exchange is not required, making the synthesis route very simple and easy to handle. Furthermore, the polymerization can be performed in any vessel by a simple heat treatment, enabling the production of porous resorcinol–formaldehyde resin monoliths (PRFM) in any shape.

Received: April 12, 2019

Accepted: June 12, 2019

Published: June 12, 2019

The possibility of synthesizing a porous monolith with a monomodal pore size distribution made this material ideal for the characterization of confined solutions. Especially, understanding ionic motion in a confined space is crucial for use in lithium-ion batteries, as one approach to overcome the issue of leakage in lithium-ion batteries is the incorporation of the electrolyte, ionic liquid (IL),^{17–19} as well as carbonate or other organic solvent-based electrolytes,^{20,21} in a nanoporous material. Ionic liquids are particularly attractive since they exhibit negligible vapor pressures, electrochemical and thermal stability, and nonflammability. In addition, the pore wall chemistry of the confining material can be tuned for optimal interactions between the pore walls and the different ion species, enabling selective control of ionic motion according to their charges as desired. This advantage can be utilized to enhance the transference number of Li^+ in battery application or to tune the thermoelectric properties, possessing either a positive or negative Seebeck coefficient.^{22,23}

So far, ionic liquids have been confined mainly in nanoporous silica^{18,19,24,25} or carbon^{26–28} hosts. There are several examples of the dynamic properties increasing^{19,29} as well as decreasing^{30,31} when an IL is confined, showing that the effect is dependent on the properties of the IL³² and the confining space^{33,34} at a molecular level. Furthermore, it was found that the loading fraction also plays a vital role, as the dynamics of the IL close to the pore walls is different from the more bulklike behavior at the pore center.^{32,35,36} But very little is known about the influence of the pore diameter on the ionic conductivity. Ionic conductivities that are comparable to those in commercial separators²¹ and in the bulk state^{19,37} and even an order of magnitude higher³⁸ were reported for nanoconfined electrolytes. Néouze et al. found that the ionic conductivity of ionic liquid confined in a mesoporous silica-like network (ionogel) increases with increasing pore size and ionic liquid content.¹⁷ A more detailed survey has been done by Guymard-Lack et al.¹⁸ who investigated ionic liquid-filled silica monoliths with a hierarchical pore system consisting of macropores of 4 μm diameters and mesopores with varying sizes ranging from 3.7 to 20 nm. It was suggested that the macroporosity does not have a beneficial effect on the ion transport but rather the mesopores increase the mobility of the confined ionic liquid, reaching a maximum ionic conductivity for a pore size of 12 nm.

Yet, a systematic study about the effect of the pore size and the surface chemistry on the ionic motion through pore network over a wide range of pore sizes is still pending, which is where this work comes in. Most nanoporous host materials, however, have the disadvantage of being powdery, resulting in a high quantity of liquid adsorbed on the particle surface or bulk liquid embedded between particles. This liquid residing outside of the pores can lead to erroneous results. Hence, in the present study, the advantage of the polymer host exhibiting a monomodal pore size distribution as well as the feasibility to form polymer into the desired shape was employed to gain a better insight into the effect of nanoconfinement on an ionic liquid-based electrolytic solution (ES). Moreover, the chemical structure of RF resin that is rich in hydroxyl and ether groups allows us to study the influence of polarity of the pore surface on the ionic motion of confined ionic liquid-based electrolyte. We show that the pore size of polymeric RF monoliths can be precisely tuned by the amount of ethanol added in the reaction mixture. 1-Methyl-1-propylpyrrolidinium bis(trifluoromethylsulfonyl)imide ($\text{Pyr13}^+\text{TFSI}^-$)-based electrolytic solution is

loaded into the porous RF networks and their thermal stability, melting behavior, and dynamic properties are examined. In particular, to investigate the ion transport through porous RF monoliths, conductivities from AC impedance spectroscopy and diffusion coefficients from pulsed field gradient (PFG) NMR spectroscopy are compared since these two values are correlated by the Einstein relation. Mass transport through three-dimensionally interconnected pore network is strongly influenced by the microstructure of pore network geometry, which is different from the mass transport in bulk. Porosity, tortuosity (ratio of the length of the pore percolation pathway over the length of the sample along the bulk flow direction), distribution of the pore diameter along the pore, constricted pore pathway etc. are important geometrical parameters. We will show that both conductivity and diffusion coefficients of electrolytic solution inside porous RF monoliths can be well described taking into account pore network structure and some geometrical parameters can be obtained. AC impedance data indicate that an IL-based electrolytic solution in RF pores provides good ionic conductivities, which depend on the pore volumes. Further, we demonstrate that cationic and anionic diffusion through the mesoporous network can be separately determined by pulsed field gradient NMR spectroscopy, providing valuable information about the interfacial interaction. We determine that the cationic diffusion of IL inside a porous structure is reduced due to the electrostatic interaction between the cations and pore surface, resulting in a composite system where IL anionic transport is faster than that of IL cations. Our results suggest that the ionic transport can be selectively tuned by the designing of the porous framework chemistry.

2. EXPERIMENTAL SECTION

2.1. Chemicals. For the synthesis of the porous RF-monolith resorcinol (Sigma Aldrich, 99%), formaldehyde (Grüssing, 37 wt % in water, stabilized with methanol), hydrochloric acid (VWR Chemicals, 37 wt % in water), and ethanol (Acros Organics, 99.5%) were used. The electrolytic solution (ES, 0.5 M) was prepared by dissolution of 1 g of bis(trifluoromethane)sulfonimide lithium salt (Li^+TFSI^- , Sigma-Aldrich, $\geq 99\%$) in 10 g of 1-methyl-1-propylpyrrolidinium bis(trifluoromethylsulfonyl)imide ($\text{Pyr13}^+\text{TFSI}^-$, IoLiTec, 99%). The chemicals were used as received without further purification.

2.2. Synthesis of Porous RF-Monoliths. The porous RF-monoliths were synthesized on the basis of the synthetic protocol of Hasegawa et al.¹² with further modifications.¹³ For the synthesis of the porous RF-monoliths, 2.00 g of resorcinol was dissolved in 3.00 mL of hydrochloric acid (0.01 M) and a varying amount of ethanol. The molar ratio of formaldehyde to resorcinol (2.2) as well as the amount of catalyst added was kept constant. The exact values are shown in Table S1. The solution was cooled in an ice bath for 15 min before addition of 3.00 mL of formaldehyde solution and then stirred at 0 °C for another 15 min. The ice cold solution was transferred in a polypropylene tube and placed in a water bath at 40 °C for 24 h for gelation. The monoliths were further polymerized in an oven at 60 °C for 24 h and afterward dried there for 10 days. In this manuscript, these porous polymer monoliths are referred to as PRFM- x , in which x specifies the volume of ethanol used and the composites of porous RF polymer monolith filled with electrolytic solution are denoted as PRFM- x -ES.

2.3. Morphology and Porosity Characterization. Nitrogen physisorption measurements were performed on a Quantachrome Quadrasorb-SI-MP and an Autosorb-6-MP instrument at 77.4 K. The PRFM-samples were crushed and degassed at 30 °C for 20 h prior to the measurement. The pore size distributions were calculated using the Barrett–Joyner–Halenda (BJH) method³⁹ as well as density functional theory (DFT)⁴⁰ models. For the DFT models, cylindrical

pores were assumed and the adsorption and desorption branches were evaluated for carbon and silica as adsorbent. The specific surface areas of the monoliths were obtained by the Brunauer–Emmett–Teller (BET)⁴¹ method. All calculations were performed using Quantachrome Instruments ASiQwin software. Scanning electron microscopy (SEM) images were taken on a GEMINI LEO 1550 microscope operated at 1 kV.

2.4. Thermal Property. Thermogravimetric analyses (TGA) was performed using a NETZSCH STA 409C with a heating rate of 5 K min^{−1} and a gas flow of 50 mL min^{−1} of pressured air. Differential scanning calorimetry (DSC) was performed using a NETZSCH DSC F1 Phoenix with a heating rate of 10 K min^{−1} from −120 to 60 °C under a nitrogen atmosphere. To prepare samples filled with electrolytic solution, the PRFM-samples were placed in an excess of the electrolytic solution in an argon glovebox for at least 12 h and the excess liquid on the monolith surface was cleaned with a paper towel. The composite samples were sealed in Al pans. After cooling to −120 °C, the samples were thermally annealed at −20 °C to achieve complete crystallization.

2.5. Solid-State NMR Spectroscopy. ¹³C cross polarization (CP) magic angle spinning (MAS) NMR spectra of porous RF monoliths were acquired at ¹³C frequency of 100.66 MHz on a Bruker Avance II 400 spectrometer, utilizing ramped polarization transfer from proton to carbon, a spinning rate of 13 kHz, and two-pulse phase-modulated decoupling during acquisition. ¹H 90° pulse length of 4.2 μs, contact time of 1 ms, and repetition delay of 4 s were used.

2.6. Conductivity Measurement. For conductivity measurements, the PRFM-samples were cut into 3–5 mm thick pellets and after drying in vacuum at 30 °C overnight, immersed in an excess of electrolytic solution for at least 24 h. Potentiostatic electrochemical impedance data were collected by a BioLogic VMP3 potentiostat using an air-tight, two-electrode (stainless steel) set from rhd instruments (TSC battery cell) over the frequency range from 1 MHz to 10 Hz with an amplitude of 50 mV. The temperature was controlled from 5 to 60 °C using a Microcell HC cell stand (rhd instruments), and the impedance data evaluation was performed using RelaxIS 3 software (rhd instruments). Apparent porosity (θ_{app}), which is a fraction of the volume of the impregnated electrolytic solution over the total sample volume, was considered to analyze conductivity on the basis of the extended 3-dim pore structural model. PRFM-samples were weighed before and after impregnation with the electrolytic solution and 1.47 and 1.25 g cm^{−3} as densities of electrolytic solution and RF polymer, respectively, were used for calculating apparent porosity.

2.7. Pulsed Field Gradient (PFG) NMR. The PFG NMR measurements of electrolytic solution confined in porous polymer monoliths were performed on a Bruker Avance III HD 600 spectrometer. The PRFM samples were cut into long rods with the dimensions of the NMR tube, and the electrolytic solution was filled into the pores by immersing the rod in an electrolytic solution overnight and wiping off the remaining solution on the outside surface of the rod. In this way of sample preparation, propagation of ES in the interparticle space can be minimized, enabling transport behavior of confined ES through co-continuous pores to be monitored. The self-diffusion coefficients of Pyr13⁺ and TFSI[−] were determined from the echo signal attenuation of the ¹H and ¹⁹F NMR spectra, respectively, acquired with a stimulated echo bipolar pulse-gradient pulse (stebpgp) sequence. The gradient strength was varied in 16 steps from 2 to 95% of the maximum gradient strength of the probe (51.3 G cm^{−1}). Diffusion coefficient *D* was extracted from the fitting of the echo signal decay to the equation

$$I = I_0 e^{-D(\gamma g \delta)^2 (\Delta - \delta/3 - \tau/2)} \quad (1)$$

where *I* is the echo signal intensity at gradient strength *g*, *I*₀ is the echo signal intensity without a gradient, γ is the gyromagnetic ratio of the nuclei, δ is the gradient pulse length, Δ is the time interval between two gradient pulses, and τ is the duration between the bipolar gradient. All measurements were carried out at 60 °C.

3. RESULTS AND DISCUSSION

3.1. Morphology and Chemical Structure of PRFM.

Porous resorcinol–formaldehyde resin monoliths (PRFM-*x*; *x* denotes volume of ethanol used for the synthesis) were prepared by sol-gel polymerization of formaldehyde and resorcinol with a constant molar ratio (formaldehyde/resorcinol = 2.2) and varying amount of ethanol as a cosolvent. SEM images of all PRFM samples show co-continuous porous structures, in which the pores appear to be irregularly formed and disordered. Selected examples are displayed in Figure 1.

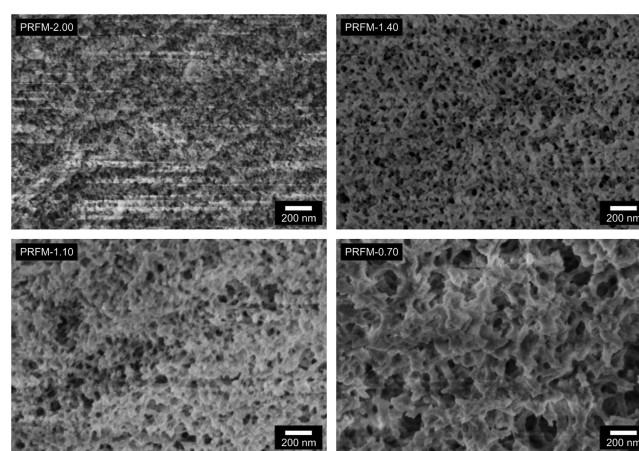


Figure 1. SEM images of selected porous resorcinol–formaldehyde resin monolith samples. The samples are denoted as PRFM-*x*, where *x* represents volume of ethanol used for the synthesis.

These examples show clearly that all samples exhibit a well-connected network of pores with the pore size decreasing with increasing amount of ethanol but the pore network topology remains unchanged.

The porosity of the PRFMs was more quantitatively characterized with nitrogen physisorption measurements (physisorption isotherms are shown in Figure 2a for selected samples and in Figure S1 for all samples). The physisorption isotherms for samples PRFM-2.00 to PRFM-1.10 show type IV(a) isotherms with H2(a) or H2(b) hysteresis loops resulting from percolation or pore-blocking/cavitation effects,⁴² which is characteristic for disordered mesoporous channels. For PRFM-0.70, a type II isotherm is seen, indicating that PRFM-0.70 is a solely macroporous system. The pore size distributions of the nitrogen physisorption measurements (see Figure 2b) were calculated from the desorption branch using a DFT kernel based on a silica material with cylindrical pores, and the maximum in the pore size distribution will be referred to as the main pore size (Table 1). As the RF-resin has a very heterogeneous surface, the results have to be viewed with caution and all methods have to be compared. Further details about pore size analysis are described in the Supporting Information. The pore size distribution and pore volume of the macroporous PRFM-0.70 could not be determined with physisorption techniques. In general, the porosity of macropores can be characterized by mercury porosimetry; however, mercury porosimetry compresses the PRFMs, resulting in erroneous pore sizes. Therefore, a main pore size for PRFM-0.70 can only be estimated as ~100–200 nm from the SEM image.

The results of the nitrogen physisorption measurements as a function of the molar ratio of ethanol to water are summarized

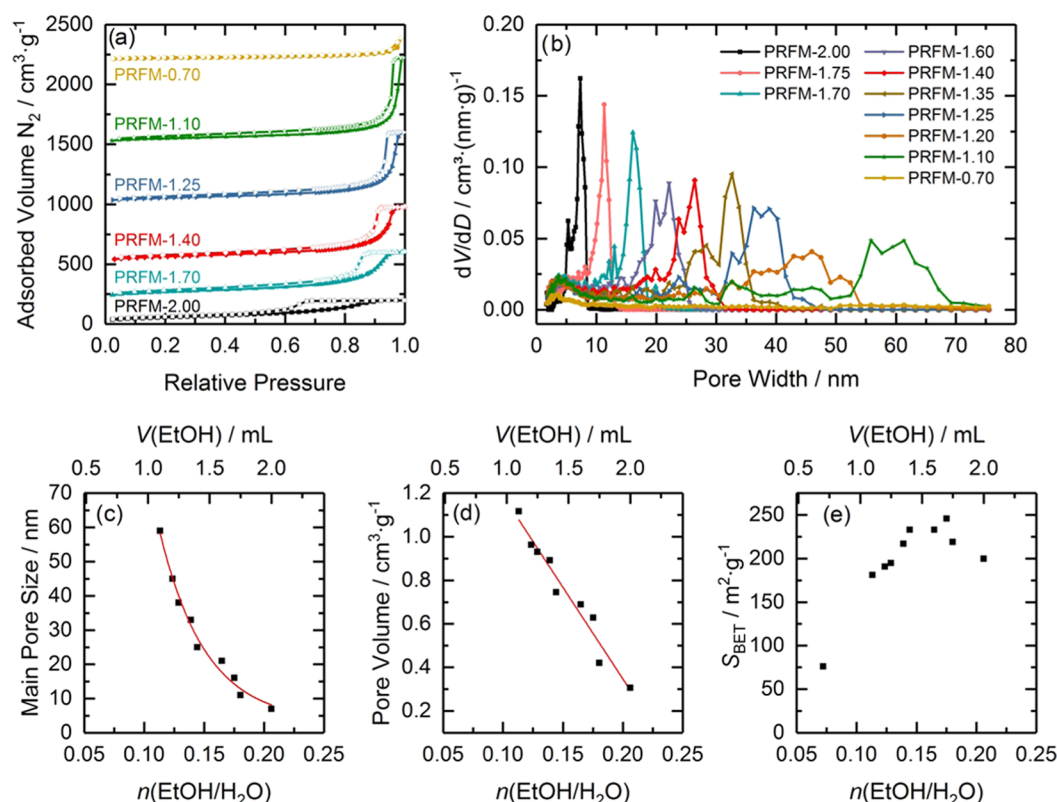


Figure 2. (a) Representative nitrogen physisorption isotherms (77 K) and (b) pore size distribution of porous RF-monoliths. Overview of (c) average pore diameter, (d) pore volume, and (e) surface area of porous RF-monoliths as a function of the molar ratio of ethanol to water used for the synthesis.

Table 1. Average Pore Diameter, Pore Volume, and Surface Area of Porous RF-Monoliths Obtained from the N_2 -Physiosorption Measurements

sample	$n(\text{EtOH})/n(\text{H}_2\text{O})$	d_{pore} (nm) ^a	V_{pore} ($\text{cm}^3 \text{g}^{-1}$)	S_{BET} ($\text{m}^2 \text{g}^{-1}$)
PRFM-2.00	0.206	7	0.31	200
PRFM-1.75	0.180	11	0.42	219
PRFM-1.70	0.175	16	0.63	246
PRFM-1.60	0.165	21	0.69	233
PRFM-1.40	0.144	25	0.74	233
PRFM-1.35	0.139	33	0.89	217
PRFM-1.25	0.129	38	0.93	195
PRFM-1.20	0.124	45	0.96	191
PRFM-1.10	0.113	59	1.12	181
PRFM-0.70	0.072	nd	nd	76

^aThe pore size stated is the maximum of the pore size distribution calculated from the desorption data using a silica equilibrium kernel.

in Figure 2c–e and Table 1. The main pore size decreases with an increasing amount of ethanol, following an exponential decay function. This is consistent with the previous report of porous carbon monoliths, showing that the average pore size is proportional to the inverse of the relative amount of ethanol when a large quantity of ethanol is used.¹³ This change in the pore size has been explained by the phase separation mechanism between a hydrophobic polymer phase and a hydrophilic solvent phase via spinodal decomposition and delayed phase separation due to the interplay of ethanol.^{12,13} Our results demonstrate that the main pore size of the PRFMs can be adjusted very precisely over the entire mesoscale

ranging from smaller than 10 nm up to macropores with pores around 60 nm and even beyond the pore size range that nitrogen physisorption can display. The pore volumes decrease linearly upon increase of $n(\text{EtOH}/\text{H}_2\text{O})$, whereas BET surface area increases with increasing $n(\text{EtOH}/\text{H}_2\text{O})$, reaching a maximum of $248 \text{ m}^2 \text{g}^{-1}$ for PRFM-1.70 and decreases afterward.

Chemical structures (i.e., the degree and type of cross-linking) of PRFMs with different pore sizes are compared by ^{13}C CP MAS NMR spectroscopy (Figure 3). For both samples, broad resonances at 152, 131, 118, 63, and 22 ppm are

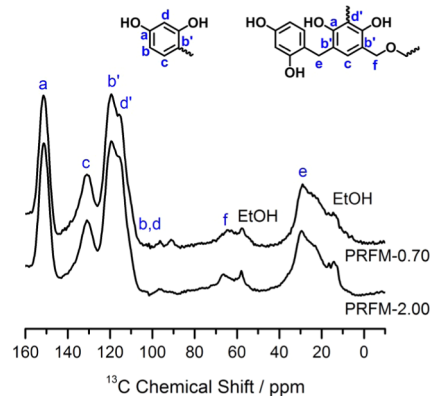


Figure 3. ^{13}C CP MAS NMR spectra of the RF-polymers with the smallest (PRFM-2.00) and the largest (PRFM-0.70) pores. Structural units of RF-polymers and the NMR signal assignments are denoted in the figure.

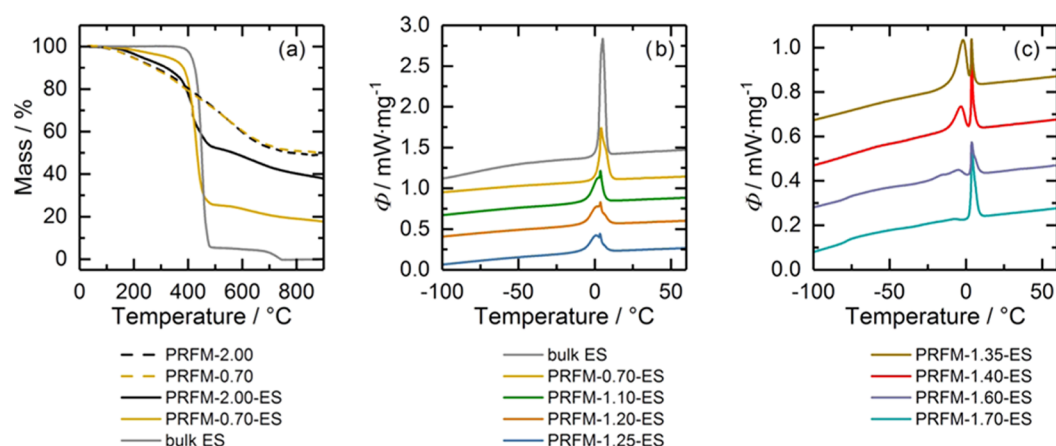


Figure 4. (a) TGA and (b, c) DSC measurements of PRFMs filled with electrolytic solution (denoted as PRFM-*x*-ES) and the bulk electrolytic solution (ES) measured in a nitrogen atmosphere.

observed, which can be assigned to aromatic carbon directly bonded to $-OH$, nonsubstituted aromatic carbon (meta to $-OH$), substituted aromatic carbon (ortho to $-OH$), methylene ether bridge ($-\text{CH}_2-\text{O}-\text{CH}_2-$), and methylene bridge ($-\text{CH}_2-$), respectively.^{43–45} The presence of the signal at 131 and 118 ppm together with the disappearance of the signal at 100 ppm (resulting from nonsubstituted aromatic carbon that is ortho to $-OH$) indicates that the polymerization occurs through the addition of formaldehyde to ortho- and para-position of resorcinol ring (2,4,6- ring position). The broad signals centered at 63 and 22 ppm imply the cross-linked polymer structure containing methylene and methyl ether bridges. Broad line widths suggest that the monoliths show low mobility and high rigidity resulting from the highly cross-linked polymeric structure as well as the structural inhomogeneity arising from mono-, di-, and tri-substituted linkage to the resorcinol ring. A similar spectral pattern of PRFM-2.00 and PRFM-0.70 suggests similar mobility and polymerization extent of both PRFMs. Therefore, assuming that both materials exhibit similar cross polarization dynamics, the degree of cross-linking is deduced from the intensity ratio of the signal at 22 ppm (methylene bridge) to the signal at 152 ppm (aromatic carbon with an OH) as an arbitrary measure for structural analogy.⁴⁶ The methylene/resorcinol ratios of ~ 0.9 were obtained for both materials, indicating that the samples with different pore sizes still have the same connectivity between the resorcinol rings and thus the same chemical composition of the pore walls. Note that this ratio does not represent quantitative value of methylene and resorcinol amounts but provides an arbitrary measure since the signal intensity of CP NMR depends on the chemical structure and the mobility of individual functional groups, yielding only semiquantitative information.

3.2. Thermal Properties of Ionic Liquid Confined in Mesopores. Thermal stability and phase behavior of confined ionic liquid are investigated with an electrolytic solution (ES) of 0.5 M bis(trifluoromethane)sulfonimide lithium salt (Li^+TFSI^-) in 1-methyl-1-propylpyrrolidinium bis-(trifluoromethylsulfonyl)imide ($\text{Pyr13}^+\text{TFSI}^-$) loaded into the pores of PRFMs with various pore sizes. The thermogravimetric analysis (Figure 4a) of the PRFM-*x* polymers (dashed lines) shows continuous mass loss from 150 to 700 °C. The mass loss at low temperatures (up to ca. 300–350 °C) can be attributed to the release of solvent and

unreacted formaldehyde entrapped in the polymer during the polymerization and formaldehyde that is being released due to rearrangements in the polymer, because the polymerization and drying were performed at only 60 °C.^{43,46} At higher temperatures, the carbonization takes place with the release of functional groups. The PRFMs containing electrolytic solution (denoted as PRFM-*x*-ES) also show a gradual mass loss between 150 and 400 °C, whereas a steep degradation step is observed at approximately 400 °C, which is correlated with the evaporation or decomposition of the ionic liquid. Afterward, the carbonization of the polymer and further decomposition of LiTFSI results in a continuous mass loss. Thus, TGA results indicate good short-term thermal stability of PRFM-*x*-ES composites up to almost 400 °C. The total mass loss can be correlated with the different ratios of electrolytic solution to polymer matrix depending on the pore volume. However, no significant difference in the temperature behavior depending on the pore size of the PRFM is seen.

To investigate the phase behavior of electrolytic solution confined in pores at thermal equilibrium condition, DSC was carried out with an annealing step at -20 °C. DSC heating curve of the bulk electrolytic solution (see Figure 4b) shows a double melting peak with an onset at 2.1 °C (peak at 5.2 °C), which is in agreement with a previous study.⁴⁷ Note that the melting point of electrolytic solution is lower than that of pure IL (mp = 10 °C, peak) owing to the salt effect.⁴⁸ It has been reported that the thermal behavior of these IL–salt mixtures is complex and sensitive to the salt concentration and thermal history, which exhibits multiple transitions due to a solid–solid phase transition and melting.^{49,50} The macroporous PRFM-0.70-ES composite shows a thermal behavior similar to that of the bulk electrolyte. In contrast, the PRFM-*x*-ES composites with mesopores exhibit additional melting peaks that occur at lower temperatures upon decreasing pore size as well as a sharp melting peak at the same temperature with the bulk electrolyte (PRFM-1.10-ES to PRFM-1.70-ES, Figure 4b,c). The sharp melting peak at approximately 2 °C is associated with ES traces on the particle surface of the PRFM-*x*-ES composite and the broad peaks at lower temperatures result from the melting behavior of ES confined inside the pores.^{49,51} For PRFM-*x*-ES composites with large mesopores (from PRFM-1.10-ES to PRFM-1.40-ES), only one broad melting peak of confined ES is seen, whereas for samples with smaller mesopore sizes (PRFM-1.70-ES and PRFM-1.60-ES), two melting peaks of

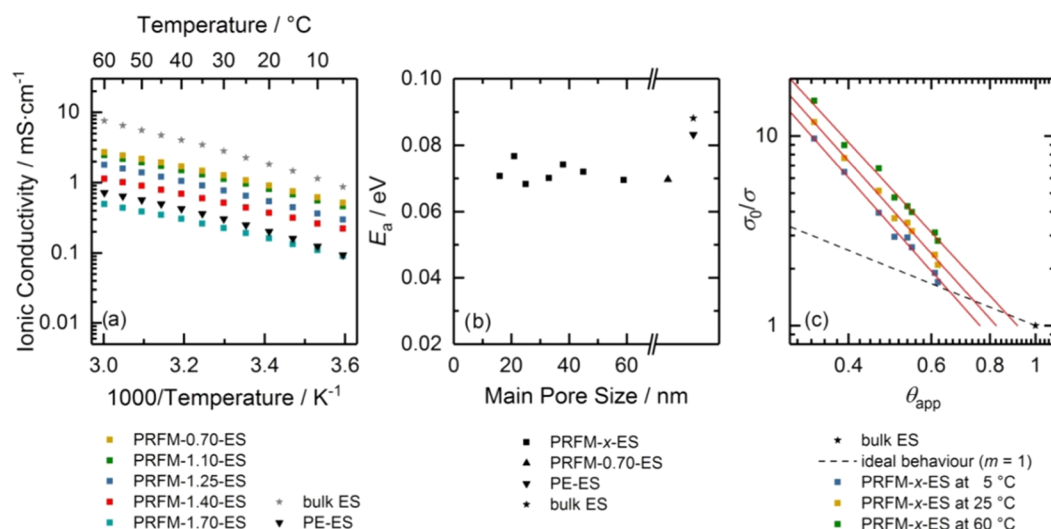


Figure 5. (a) Ionic conductivities of the PRFMs filled with electrolytic solution at various temperatures. (b) Activation energy for ionic conduction of PRFM-*x*-ES as a function of pore diameters. Commercial PE separator Celgard™ K2045 with the electrolytic solution and the bulk electrolytic solution were shown together as references. (c) Correlation between σ_0/σ and porosity of PRFM-*x*-ES at 5, 25, and 60 °C, following Archie's formulation represented by the solid line. The dashed line represents the ion transport behavior inside uniform cylindrical pores of equal diameters that are aligned parallel to the electric field (tortuosity = 1) with neither interconnection nor bottleneck. The point for the bulk electrolytic solution is denoted as a star symbol.

confined ES are visible as well as glass transition at -75 °C. It is likely that for smaller pores, ES in contact with the pore surface (adsorbed layer) becomes more dominating than those in the center of the pores (bulklike layer) due to the smaller pore sizes or partial pore filling, exhibiting two melting peaks at slightly different temperatures. Dynamic and thermodynamic heterogeneity of the confined species have been observed in previous studies.^{34,35} Likely, changes in ES–surface interaction cause a different response of ES to temperature variation, delaying crystallization of the ES and leading to the metastable phase that undergoes solid–solid phase transition. Similar glass transition and two exothermic peaks are observed when the DSC was performed without any annealing step at low temperatures (see Figure S4). The extent of melting point depression is inversely proportional to the pore radius, lowering the melting point down to -10 °C at the pore radius of 16 nm when the peak temperature in DSC is defined as a melting point (Figure S5). When the onset temperature of the melting point is used instead (Figure S5), the melting point depression exhibits a linear trend as a function of inverse pore radius up to pore sizes of 25 nm (PRFM-0.70-ES to PRFM-1.40-ES) and deviates from the linearity for the pore sizes smaller than 25 nm (PRFM-1.60-ES and PRFM-1.70-ES). This deviation is due to the second weak and broad melting peak, which arises from the ES in the adsorbed layer. Nonetheless, the PRFM-*x*-ES composites show reduction in the melting point following the Gibbs–Thomson equation,^{52,53} which can broaden the temperature window of the ES and thus can be utilized for low-temperature application.

For samples with even smaller pores (PRFM-2.00-ES and PRFM-1.75-ES, Figure S6), only a melting peak of bulk electrolytic solution (above 0 °C) is observed, suggesting that there is not enough electrolytic solution in the pores for detection with the DSC. It has been shown that the small mesopores cannot be filled with ionic liquid at atmospheric pressures.⁴⁹ Therefore, these composite samples with incomplete pore filling will not be considered for further study.

3.3. Ion Transport of Ionic Liquid Confined in Mesopores: Conductivity.

Ionic motion of confined electrolytic solution was studied by measuring ionic conductivities of PRFM-*x*-ES composites as well as of bulk ES and a polyethylene separator–ES composite (PE–ES, with a Celgard K2045 separator) at various temperatures, as shown in Figure 5a. Since the monoliths filled with ES were cut into disks for the conductivity measurement, the influence of the ES residing on the particle surface is negligible. Over the whole temperature range, bulk ES shows the highest conductivity and the conductivities of the PRFM-*x*-ES composites vary depending on the pore sizes. The PE–ES composite exhibits lower conductivity than most of the PRFM-*x*-ES composites (except PRFM-1.70-ES). It can be seen that the ionic conductivities of PRFM-*x*-ES composites increase with increasing pore sizes and volumes, which is not surprising since conductivity is proportional to the number of mobile charge carriers ($\sigma_0 = nq\mu$; n = effective number density of charge carriers, q = the elementary electric charge, μ = the mobility of charge carrier); higher conductivity is expected for composites with larger pore volumes.

Thus, to get information about the confinement effect on the ion transport motion, the temperature dependence of the conductivity is examined since activation energy can be used as an indicator for the ionic mobility. All samples show that the thermal activation process of these materials can be well described by the Vogel–Fulcher–Tamman (VFT) equation, which is typically seen for the glass-forming liquids.³³

$$\sigma = \frac{A}{\sqrt{T}} e^{-E_a/k_B(T-T_0)} \quad (2)$$

where A is the pre-exponential factor, E_a is the apparent activation energy, k_B is the Boltzmann constant, and T_0 ($T_0 \approx T_g - 50$ K) is the ideal glass transition temperature.⁵⁴ The apparent activation energies are extracted from the fitting of $\ln(\sigma T^{0.5})$ against $1/(T - T_0)$. All PRFM-*x*-ES composite materials exhibit slightly lower apparent activation energies than the bulk electrolytic solution and the PE–ES composite,

with no notable dependency on their pore diameter (Figure 5b). The data were also analyzed on the basis of the Arrhenius equation, yielding activation energies similar to those from VFT analysis; however, the fitting does not match the measured values as well as the VFT fits (Figure S7). At first glance, this slight reduction in apparent activation energy by confinement appears to be contradictory since slower motion of ionic species or increased tortuosity of conducting phase is often associated with higher activation energies.^{55,56} Possible reasons for slight differences in E_a between bulk electrolyte and PRFM-*x*-ES composites can be changes in viscosity or in configuration of the ions upon confinement. Additionally, interactions of the pore walls with the different ionic species may lead to better dissociation of ion pairs, lowering E_a . Different results from the literature (increase^{55,56} or decrease^{19,57} of E_a upon confinement) suggest that the interactions of the pore walls with the different ionic species play an important role as well as the geometrical confinement itself. Yet, by only considering the apparent activation energies, the physicochemical picture for the ionic motion can only be speculated. Nonetheless, E_a values similar to those of the PE-ES composite demonstrate potential applicability of PRFM-*x*-ES composite materials in an energy storage device. Intriguingly, no clear dependence of E_a (ionic mobility) on the pore diameter can be recognized, which is contrary to the results of Guyomard-Lack et al. who identified the maximum ionic mobility at approximately 10 nm pores.¹⁸ It appears that the ionic conductivity measurement is less sensitive to the confinement and may not be a good measure for the effect of the nanoconfinement on the ionic transport motion since it probes collective dynamical properties rather than the motion of individual ions.^{24,32}

Mass transport through extended pore network can be described by the empirical formulation by Archie

$$\frac{\sigma_0}{\sigma} = \theta^{-m} = F \quad (3)$$

where σ and σ_0 are conductivities of liquid in porous media and bulk liquid, respectively, θ is the porosity fraction of porous media, and F is defined as the formation factor.^{58,59} For various sands filled with electrolyte, Archie observed a linear correlation between $\log F$ and $\log \theta$ with slope $-m$, where m reflects the effect of the microstructure of porous media, such as tortuosity, variation of pore diameters, constrictivity, and bottleneck. Since RF monoliths are porous media containing three-dimensional co-continuous pores with pore size distribution along the percolating path, we examined whether Archie's law can be applied to explain the ionic conductivity behavior of PRFM-*x*-ES composites. Because small pores cannot be completely filled with relatively large liquid molecules, apparent porosity (θ_{app}) that is a ratio of pore volumes filled with electrolytic solution over total sample volumes is utilized instead of porosity obtained by gas physisorption measurement. As shown in a logarithmic plot of σ_0/σ vs θ_{app} (Figure 5c), PRFM-*x*-ES composites exhibit a linear trend with $m = 2.8$ at all temperature ranges studied, following Archie's law. This indicates that all RF monoliths display the same structural parameters of the pore network, which is consistent with SEM images and for such a series of porous media with similar structures, the ionic conductivity depends on porosity. This Archie's behavior suggests that not only the pore diameter but also various geometrical aspects of pore structure need to be considered to explain the mass

transport properties inside a co-continuous pore network structure. However, a difference from the ideal Archie's formulation is noted. The straight line with a slope $-m$ does not pass through the point of the electrolyte solution ($\sigma_0/\sigma = 1$ at $\theta_{app} = 1$). This nonideal Archie's behavior can be seen more clearly in a plot of σ vs θ_{app} , where extrapolation of the θ^m power function to the point of bulk electrolytic solution shows a strong deviation from the bulk electrolytic solution (Figure S8). It appears that the underlying mechanism of charge transport is more complex and cannot be explained simply by considering only the influence of geometrical parameters. The surface chemistry of the material seems to have a large effect. Since contribution of individual factors to the confinement effect cannot be quantitatively acquired from conductivity measurement, another technique to investigate mass transport is necessary, which will be described below.

3.4. Ion Transport of Ionic Liquid Confined in Mesopores: Diffusion Coefficients. To get a detailed insight into the influence of the geometric confinement and the surface chemistry of the pore walls on the mass transport, time-dependent diffusion coefficients of the ionic liquid inside mesopores were determined using PFG NMR spectroscopy. At short diffusion times (Δ) when the spins are allowed to travel for a short time, the macroscopic morphology of the porous media has little influence on the diffusion and whether the diffusion process occurs inside closed pores or connected open porous network cannot be distinguished. The short time diffusion coefficient depends only on the surface-to-volume ratio (S/V) and approaches the bulk diffusion coefficient (D_0) of unrestricted spins at $\Delta = 0$ s. As diffusion time increases, the spins will migrate to longer distances, being significantly influenced by the restricted geometry, and the diffusion coefficients decrease asymptotically, reaching constant values (steady-state diffusion coefficient, D_∞) at long diffusion time. Bulk diffusion coefficients (D_0) and surface-to-volume ratio (S/V) can be extracted from the extrapolation to $\Delta = 0$ s and the slope, respectively, according to the Mitra equation (eq 4) in the short time region (see Figure S9a).^{60,61} Tortuosity (α , eq 5), representing the degree of the tortuous structure, can be deduced from the limiting values at long diffusion time and bulk diffusivity (D_{bulk}).^{35,60}

$$D(\Delta) = D_0 \left(1 - \frac{4S}{9\sqrt{\pi V}} \sqrt{D_0 \Delta} \right) \quad (4)$$

$$\alpha = \frac{D_{bulk}}{D_\infty} \quad (5)$$

S/V , which is a measure of the microscopic length scale, plays a key role in systems where the surface interaction drives chemistry, whereas α , which is a geometrical parameter on a macroscopic length scale, reflects porous network geometry that is an important factor in various transport processes. Note that α here is a diffusive tortuosity, capturing all microstructural effects (bottleneck, dead end, etc.) and interfacial interaction that influence the diffusivity rather than pure geometric tortuosity. Thus, valuable information about the surface interaction and geometry of porous media can be obtained from a series of diffusion measurements at various times.

IL confined in the porous RF-polymer monoliths still exhibits a liquidlike character and its diffusion behavior can be investigated by PFG NMR spectroscopy. Since NMR spec-

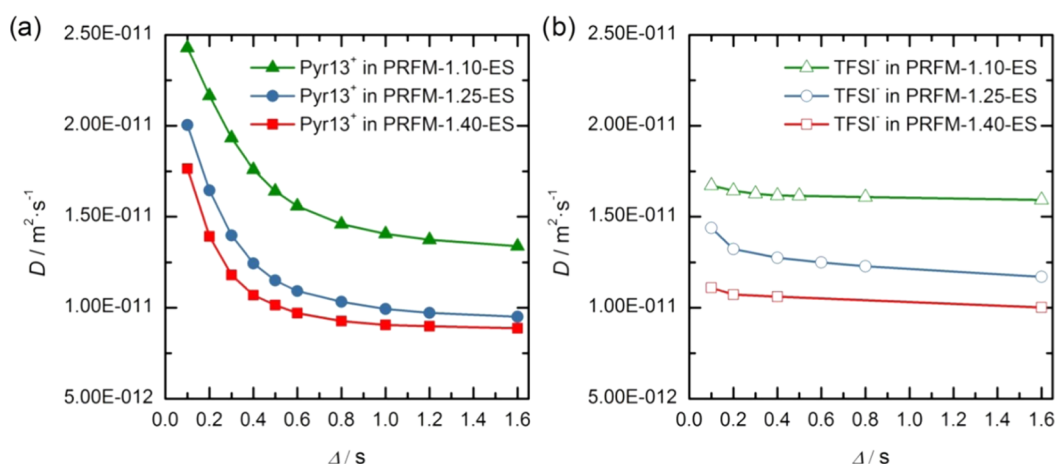


Figure 6. Diffusion coefficients of the confined (a) Pyr13⁺ cation and (b) TFSI[−] anion as a function of the diffusion time (Δ). All measurements were carried out at 60 °C.

Table 2. Estimated Values of the Bulk Diffusion Coefficient, Surface-to-Volume Ratio, Steady-State Diffusion Coefficients, and Tortuosity Parameters of PRFM-*x*-ES Composites at 60 °C^a

sample	pore diameter (nm)	D_0 ($10^{-11} \text{ m}^2 \text{ s}^{-1}$) ^b	S/V (μm^{-1}) ^b	D_{∞} of Pyr13 ⁺ ($10^{-11} \text{ m}^2 \text{ s}^{-1}$) ^c	D_{∞} of TFSI [−] ($10^{-11} \text{ m}^2 \text{ s}^{-1}$) ^d	α by Pyr13 ⁺ ^c	α by TFSI [−] ^d
PRFM-1.10-ES	59	3.0	0.47	1.3	1.6	4.1	2.3
PRFM-1.25-ES	38	2.7	0.64	0.95	1.2	5.8	3.1
PRFM-1.40-ES	25	2.4	0.74	0.89	1.0	6.2	3.6

^aDiffusion coefficients of the bulk electrolytic solution at 60 °C are $D_{\text{Pyr13, bulk}} = 5.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, $D_{\text{TFSI, bulk}} = 3.6 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, and $D_{\text{Li, bulk}} = 2.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$. ^bThese values are extracted for Pyr13⁺ cations since only ¹H PFG NMR experiments exhibit a time-dependent diffusion behavior. ^cSteady-state diffusion coefficients and tortuosity of Pyr13⁺ cation are obtained from ¹H PFG NMR data. ^dSteady-state diffusion coefficients and tortuosity of TFSI[−] anion are obtained from ¹⁹F PFG NMR data.

troscopy can probe individual nuclei, time-dependent diffusion coefficients of Pyr13⁺ cations and TFSI[−] anions were separately measured from ¹H and ¹⁹F NMR spectroscopy at 60 °C, as shown in Figure 6. Diffusion coefficients of Li cations could not be measured due to the short relaxation times of ⁷Li and limitation in the gradient strength. Here, it is worthwhile to mention that the diffusion time Δ represents the length scale over which a molecule travels, which is defined as diffusion length = $(2D_0\Delta)^{1/2}$ (Figure S9b). In our study, microstructural features of a pore network in the range of 3–14 μm can be probed. Several intriguing features can be noted. First, diffusion coefficients of the Pyr13⁺ cations exhibit a strong dependence on diffusion time (Δ) showing asymptotic decreases, whereas diffusion coefficients of the TFSI[−] anions do not vary significantly with diffusion time. Second, steady-state diffusion coefficients (D_{∞} , approximated at $\Delta = 1.6$ s) of both Pyr13⁺ cation and TFSI[−] anion are lower than those in the bulk unrestricted state ($D_{\text{Pyr13, bulk}} = 5.5 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$, $D_{\text{TFSI, bulk}} = 3.6 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$) by 2–5-fold. Third, D_{∞} of the TFSI[−] anion is significantly higher than that of the Pyr13⁺ cation, which is contrary to the diffusion coefficients in the bulk state. Fourth, diffusion coefficients of both cations and anions depend on the pore size of the confining polymer, decreasing as the pore diameter decreases. A time-independent diffusion behavior (typical for bulk liquid) has been widely observed since the observable length scale of nuclear migration, limited by the specific experimental condition, is too large in comparison with the size of the cavity dimension of the porous media. Here, it is intriguing that Pyr13⁺ cation and TFSI[−] anion present different time dependencies of diffusion when they are confined in the connected porous

structure of an RF-polymer. This can be attributed to the different diffusion behavior and pathway between cations and anions, i.e., more restricted diffusion for cations interacting strongly with the pore walls and diffusion of anions traveling more freely with no significant interaction with the pore walls. In addition, cationic and anionic diffusivities cross over at long diffusion times, leading to a higher diffusion coefficient for anion than cation. This contradictory behavior is likely associated with the difference in the interfacial interaction of cations and anions inside pores. For bulk IL, faster dynamics of Pyr13⁺ cations (MW = 128.239) than that of TFSI[−] anions (MW = 280.135) is in agreement with the previous studies, which ascribe this difference in motion to the different electrostatic potential of molecules.^{62–64} Due to the negative charge on N, F, and O atoms and positive charge on C and S atoms, the TFSI[−] anions exhibit a heterogeneous electrostatic field and thus interact strongly with the surroundings, leading to the slower motion.³² In contrast, Pyr13⁺ cations interact weakly with the environments and diffuse faster, owing to their homogeneous field. This difference in transport properties between IL cations and anions is correlated with local nanostructuring, driven by the molecular structure of individual cations and anions. For IL composed of cations with long alkyl chains, slower dynamics for cations than anions has been observed, which has been attributed to the enhanced cationic network.^{63,65} When ionic liquids are confined in pores, interfacial interaction at the liquid-solid phase boundary will cause significant changes in the local structure and physicochemical properties of ionic liquids. Because resorcinol–formaldehyde resin contains a significant amount of atomic partial charges due to the ether linkages and phenolic

hydroxyl groups, electrostatic potential at the pore wall surface will play a critical role. It appears that a strong Coulombic attraction between Pyr13^+ cations and the pore wall enables the formation of an adsorbed layer where Pyr13^+ cations are densely packed in contact with the pore surface. The TFSI^- anion layer will reside at a further distance from the pore surface with increased free volumes. This change in the nanodomain structuring from a bulk (three-dimensional) to a 2-dimensional layered structure causes slowing down of the dynamics for Pyr13^+ cations to a greater extent than that for TFSI^- anions. Our observation is consistent with previous molecular dynamics (MD) simulation studies, which indicate that an attractive potential profile for cations and repulsion for anions at the silica interface results in layering of fluids where cations reside near the pore surface with higher density and anions are located far from the pore surface with lower density.^{24,34} In addition, strong interaction and close proximity of IL cations near silica surface in contrast to the anions located toward the pore center have been reported by solid-state NMR study.³⁰

Geometric parameters according to eqs 4 and 5 are listed in Table 2. Steady-state diffusivity (D_∞) of both cations and anions decreases as the pore diameter decreases, as shown in Figure S10, which reflects the geometrical confinement effect. Moreover, Pyr13^+ cations yield higher diffusive tortuosities (α) than TFSI^- anions, implying that the cations interact more strongly with the pore surface than anions due to the strong electrostatic interaction between cations and pore wall. Diffusive tortuosities increase upon decrease of the pore diameter, indicating that ions interact more with the pore wall inside smaller mesopores, which is expected considering the surface areas of the materials. Since a time-dependent diffusion behavior can be seen only for Pyr13^+ cations under our experimental conditions, D_0 and S/V are extracted for cations only (Figure S9a). Extrapolated to $\Delta = 0$ s, the diffusion coefficient D_0 is expected to approach the bulk diffusion coefficient.^{35,60,66} However, smaller D_0 values than those of bulk IL are obtained, which indicates the tortuous cationic transfer path interacting with the pore surface in microcavities. The surface-to-volume ratio increases linearly as the pore diameter decreases, which is in agreement with the physisorption measurements. However, S/V ratios obtained from these two methods are significantly different, i.e., submicron scale from diffusion measurement and sub-nanometer scale from physisorption measurement. This discrepancy may not be surprising since PFG-NMR and physisorption measurements probe different physical phenomena of different length scales. Physisorption measurements employ gas molecules that interact with the dry polymer surface, and the parameters obtainable depend on the capillary condensation, size of gas molecules, etc. On the contrary, the PFG NMR experiments can represent a more realistic S/V ratio for the fluid flow in the well-connected open porous systems, due to their sensitivity to the wetted surface and to the micrometer range length scale.⁶⁷

Overall, the influence of the pore surface chemistry as well as the pore size on the ion transport motion can be observed for IL inside porous RF-polymers by PFG NMR measurements. In the bulk state or at short diffusion times, the Pyr13^+ cations diffuse faster than the TFSI^- anions. At long diffusion times, as the molecules can diffuse up to a long distance and are significantly influenced by the restriction of the pore structure, the electrostatic interactions of the Pyr13^+ cation with the

phenolic hydroxyl groups of the pore walls become dominant, reducing the motion of the Pyr13^+ cation more than that of the TFSI^- anion. Eventually, Pyr13^+ cations diffuse slower than TFSI^- anions. This diffusion phenomenon suggests layer structuring of IL inside RF-polymer pores, where the adsorption layer is rich in cations, directly interacting with the pore wall surface and the anions are accumulated in the core of the pores, forming a bulklike layer, as shown in Figure 7. Immobilization of Li^+ via coordinating to the hydroxyl

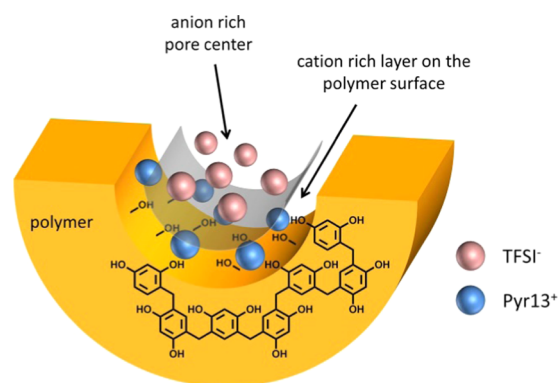


Figure 7. Schematic representation of interfacial interaction and layer structure of IL inside RF-polymer pores.

groups of the RF-polymer matrix is plausible; however, our preliminary NMR results suggest that the amount of immobile Li^+ cations is not significant (Figure S11). Presumably, Li^+ cations move together with TFSI^- anions as a complex since Li^+ is known to form solvated species with polar solvents. Further study is necessary to determine the diffusion behavior and local position of Li^+ inside the pore.

4. CONCLUSIONS

We show that the resorcinol–formaldehyde polymer monoliths with well-connected bicontinuous pore structures can be prepared exhibiting good thermal and chemical stabilities, of which pore sizes can be finely controlled by the ethanol/water ratio. This tunability and monomodal distribution of pore sizes enabled us to investigate the effect of the nanoconfinement on physical properties of an ionic liquid-based electrolytic solution in continuous open pore systems. The melting point of the $\text{Pyr13}^+\text{TFSI}^-$ -based electrolytic solution decreases as the pore size decreases, expanding the temperature window for IL to be utilized in energy storage systems. Dynamic characteristics of ions are examined by conductivity and diffusion coefficient measurements, taking into account pore network geometry. Ionic conductivity increases as the pore volume increases due to higher concentration of charge carriers. Thermal activation energy for ionic conduction is only slightly smaller than the bulk electrolytic solution; however, no clear dependence on the pore size has been observed. PFG NMR measurements reveal that the transport properties of the different ionic species of a confined IL are dependent on the surface chemistry as well as the pore sizes of the host system. Diffusion coefficients of both cations and anions decrease as the pore size decreases. In particular, the diffusion coefficient of the TFSI^- anion moving through continuous pores becomes significantly higher than that of the Pyr13^+ cation, which is contrary to the bulk state. This is due to the strong Coulombic attraction between Pyr13^+ cations and the pore wall, which is rich in

phenolic hydroxyl group and ether linkage. Different electrostatic interactions of the pore surface with cations and anions induce layering of IL with different distribution of ions inside pores and thus slow down the cationic and anionic motion to a different extent. Our results suggest that the transport characteristics of the ions of interest can be tuned by modifying the surface chemistry of the host network. Possibly, diffusion coefficients of either cations or anions can be selectively enhanced or reduced by controlling the surface hydrophilicity and solvophilicity of the host materials. Further studies are on the way to elucidate how this can be realized. In addition, we show that information about the microstructural parameter of a co-continuous porous network can be extracted from conductivity and diffusion measurements.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acsami.9b06445.

N₂ physisorption isotherms of all porous RF monoliths samples, analysis of pore size distribution using various methods, DSC analysis of PRFM samples containing ES solution, ionic conductivity analysis of PRFM-*x*-ES samples, diffusion coefficient analysis of PRFM-*x*-ES samples, fitting to the Mitra equation, and ⁷Li-MAS-NMR spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

*E-mail: lee@chemie.uni-hamburg.de (Y.J.L.).

*E-mail: michael.froebe@chemie.uni-hamburg.de (M.F.).

ORCID

Young Joo Lee: 0000-0002-5782-6431

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We thank Robert Schön for SEM measurements and Sandra König for physisorption measurements.

■ REFERENCES

- (1) Pekala, R. W. Low Density, Resorcinol–Formaldehyde Aerogels. U.S. Patent US4873218A1989.
- (2) Heidari, B. S.; Cheraghchi, V.-S.; Motahari, S.; Motlagh, G. H.; Davachi, S. M. Optimized Mercapto-Modified Resorcinol Formaldehyde Xerogel for Adsorption of Lead and Copper Ions from Aqueous Solutions. *J. Sol-Gel Sci. Technol.* **2018**, *88*, 236–248.
- (3) Wang, S.; Xu, Y.; Yan, M.; Zhang, L.; Liu, Z. The Effect of Surfactants on Carbon Xerogel Structure and CO₂ Capture. *J. Non-Cryst. Solids* **2018**, *499*, 101–106.
- (4) Awadallah-F, A.; Al-Muhtaseb, S. A. Gas Templating of Resorcinol–Formaldehyde Xerogels. *J. Non. Cryst. Solids* **2018**, *498*, 64–71.
- (5) Pahl, R.; Bonse, U.; Pekala, R. W.; Kinney, J. H. SAXS Investigations on Organic Aerogels. *J. Appl. Crystallogr.* **1991**, *24*, 771–776.
- (6) Pekala, R. W. Organic Aerogels from the Polycondensation of Resorcinol with Formaldehyde. *J. Mater. Sci.* **1989**, *24*, 3221–3227.
- (7) Rey-Raap, N.; Angel Menéndez, J.; Arenillas, A. RF Xerogels with Tailored Porosity over the Entire Nanoscale. *Microporous Mesoporous Mater.* **2014**, *195*, 266–275.
- (8) Rey-Raap, N.; Angel Menéndez, J.; Arenillas, A. Simultaneous Adjustment of the Main Chemical Variables to Fine-Tune the Porosity of Carbon Xerogels. *Carbon* **2014**, *78*, 490–499.
- (9) Hasegawa, G.; Deguchi, T.; Kanamori, K.; Kobayashi, Y.; Kageyama, H.; Abe, T.; Nakanishi, K. High-Level Doping of Nitrogen, Phosphorus, and Sulfur into Activated Carbon Monoliths and Their Electrochemical Capacitances. *Chem. Mater.* **2015**, *27*, 4703–4712.
- (10) Alonso-Buenaposada, I. D.; Rey-Raap, N.; Calvo, E. G.; Angel Menéndez, J.; Arenillas, A. Effect of Methanol Content in Commercial Formaldehyde Solutions on the Porosity of RF Carbon Xerogels. *J. Non-Cryst. Solids* **2015**, *426*, 13–18.
- (11) Alonso-Buenaposada, I. D.; Garrido, L.; Montes-Morán, M. A.; Menéndez, J. A.; Arenillas, A. An Underrated Variable Essential for Tailoring the Structure of Xerogel: The Methanol Content of Commercial Formaldehyde Solutions. *J. Sol-Gel Sci. Technol.* **2017**, *83*, 478–488.
- (12) Hasegawa, G.; Kanamori, K.; Nakanishi, K. Facile Preparation of Macroporous Graphitized Carbon Monoliths from Iron-Containing Resorcinol–Formaldehyde Gels. *Mater. Lett.* **2012**, *76*, 1–4.
- (13) Juhl, A. C.; Elverfeldt, C.-P.; Hoffmann, F.; Fröba, M. Porous Carbon Monoliths with Pore Sizes Adjustable between 10 nm and 2 μm Prepared by Phase Separation – New Insights in the Relation between Synthesis Composition and Resulting Structure. *Microporous Mesoporous Mater.* **2018**, *255*, 271–280.
- (14) Kim, S. Y.; Yeo, D. H.; Lim, J. W.; Yoo, K.-P.; Lee, K. H.; Kim, H. Synthesis and Characterization of Resorcinol–Formaldehyde Organic Aerogel. *J. Chem. Eng. Jpn.* **2001**, *34*, 216–220.
- (15) Schwan, M.; Ratke, L. Flexibilisation of Resorcinol–Formaldehyde Aerogels. *J. Mater. Chem. A* **2013**, *1*, 13462–13468.
- (16) Rey-Raap, N.; Piedboeuf, M.-L. C.; Arenillas, A.; Menéndez, J. A.; Léonard, A. F.; Job, N. Aqueous and Organic Inks of Carbon Xerogels as Models for Studying the Role of Porosity in Lithium-Ion Battery Electrodes. *Mater. Des.* **2016**, *109*, 282–288.
- (17) Néouze, M.-A.; Le Bideau, J.; Gaveau, P.; Bellayer, S.; Vioux, A. Ionogels, New Materials Arising from the Confinement of Ionic Liquids within Silica-Derived Networks. *Chem. Mater.* **2006**, *18*, 3931–3936.
- (18) Guyomard-Lack, A.; Said, B.; Dupré, N.; Galarneau, A.; Le Bideau, J. Enhancement of Lithium Transport by Controlling the Mesoporosity of Silica Monoliths Filled by Ionic Liquids. *New J. Chem.* **2016**, *40*, 4269–4276.
- (19) Iacob, C.; Sangoro, J. R.; Kipnusu, W. K.; Valiullin, R.; Kärger, J.; Kremer, F. Enhanced Charge Transport in Nano-Confined Ionic Liquids. *Soft Matter* **2012**, *8*, 289–293.
- (20) Wan, J.; Zhang, J.; Yu, J.; Zhang, J. Cellulose Aerogel Membranes with a Tunable Nanoporous Network as a Matrix of Gel Polymer Electrolytes for Safer Lithium-Ion Batteries. *ACS Appl. Mater. Interfaces* **2017**, *9*, 24591–24599.
- (21) Sakakibara, K.; Kagata, H.; Ishizuka, N.; Sato, T.; Tsujii, Y. Fabrication of Surface Skinless Membranes of Epoxy Resin-Based Mesoporous Monoliths toward Advanced Separators for Lithium Ion Batteries. *J. Mater. Chem. A* **2017**, *5*, 6866–6873.
- (22) Zhao, D.; Martinelli, A.; Willfahrt, A.; Fischer, T.; Bernin, D.; Khan, Z. U.; Shahi, M.; Brill, J.; Jonsson, M. P.; Fabiano, S.; Crispin, X. Polymer Gels with Tunable Ionic Seebeck Coefficient for Ultra-Sensitive Printed Thermopiles. *Nat. Commun.* **2019**, *10*, No. 1093.
- (23) Lee, D.; Jung, H. Y.; Park, M. J. Solid-State Polymer Electrolytes Based on AB₃-Type Miktoarm Star Copolymers. *ACS Macro Lett.* **2018**, *7*, 1046–1050.
- (24) Ori, G.; Villemot, F.; Viau, L.; Vioux, A.; Coasne, B. Ionic Liquid Confined in Silica Nanopores: Molecular Dynamics in the Isobaric–Isothermal Ensemble. *Mol. Phys.* **2014**, *112*, 1350–1361.
- (25) Li, X.; Zhang, Z.; Yin, K.; Yang, L.; Tachibana, K.; Hirano, S. Mesoporous Silica/Ionic Liquid Quasi-Solid-State Electrolytes and Their Application in Lithium Metal Batteries. *J. Power Sources* **2015**, *278*, 128–132.
- (26) Berrod, Q.; Ferdeghini, F.; Judeinstein, P.; Genevaz, N.; Ramos, R.; Fournier, A.; Dijon, J.; Ollivier, J.; Rols, S.; Yu, D.; Mole, R. A.;

Zanotti, J.-M. Enhanced Ionic Liquid Mobility Induced by Confinement in 1D CNT Membranes. *Nanoscale* **2016**, *8*, 7845–7848.

(27) Weingarth, D.; Drumm, R.; Foelske-Schmitz, A.; Kötz, R.; Presser, V. An Electrochemical in Situ Study of Freezing and Thawing of Ionic Liquids in Carbon Nanopores. *Phys. Chem. Chem. Phys.* **2014**, *16*, 21219–21224.

(28) Monk, J.; Singh, R.; Hung, F. R. Effects of Pore Size and Pore Loading on the Properties of Ionic Liquids Confined Inside Nanoporous CMK-3 Carbon Materials. *J. Phys. Chem. C* **2011**, *115*, 3034–3042.

(29) Pinilla, C.; Del Pópolo, M. G.; Lynden-Bell, R. M.; Kohanoff, J. Structure and Dynamics of a Confined Ionic Liquid. Topics of Relevance to Dye-Sensitized Solar Cells. *J. Phys. Chem. B* **2005**, *109*, 17922–17927.

(30) Nayeri, M.; Aronson, M. T.; Bernin, D.; Chmelka, B. F.; Martinelli, A. Surface Effects on the Structure and Mobility of the Ionic Liquid C₆C₁ImTFSI in Silica Gels. *Soft Matter* **2014**, *10*, 5618–5627.

(31) Waechter, M.; Sellin, M.; Stark, A.; Akcakayiran, D.; Findenegg, G.; Gruenberg, A.; Breitzke, H.; Buntkowsky, G. ²H and ¹⁹F Solid-State NMR Studies of the Ionic Liquid [C₂Py][BTA]-d₁₀ Confined in Mesoporous Silica Materials. *Phys. Chem. Chem. Phys.* **2010**, *12*, 11371–11379.

(32) Coasne, B.; Viau, L.; Vioux, A. Loading-Controlled Stiffening in Nanoconfined Ionic Liquids. *J. Phys. Chem. Lett.* **2011**, *2*, 1150–1154.

(33) Iacob, C.; Sangoro, J. R.; Papadopoulos, P.; Schubert, T.; Naumov, S.; Valiullin, R.; Kärger, J.; Kremer, F. Charge Transport and Diffusion of Ionic Liquids in Nanoporous Silica Membranes. *Phys. Chem. Phys.* **2010**, *12*, 13798–13803.

(34) Li, S.; Han, K. S.; Feng, G.; Hagaman, E. W.; Vlcek, L.; Cummings, P. T. Dynamic and Structural Properties of Room-Temperature Ionic Liquids near Silica and Carbon Surfaces. *Langmuir* **2013**, *29*, 9744–9749.

(35) Han, K. S.; Wang, X.; Dai, S.; Hagaman, E. W. Distribution of 1-Butyl-3-Methylimidazolium Bistrifluoromethylsulfonimide in Mesoporous Silica as a Function of Pore Filling. *J. Phys. Chem. C* **2013**, *117*, 15754–15762.

(36) Tripathi, A. K.; Singh, R. K. Interface and Core Relaxation Dynamics of IL Molecules in Nanopores of Ordered Mesoporous MCM-41: A Dielectric Spectroscopy Study. *RSC Adv.* **2016**, *6*, 45147–45157.

(37) Tu, Z.; Zachman, M. J.; Choudhury, S.; Wei, S.; Ma, L.; Yang, Y.; Kourkoutis, L. F.; Archer, L. A. Nanoporous Hybrid Electrolytes for High-Energy Batteries Based on Reactive Metal Anodes. *Adv. Energy Mater.* **2017**, *7*, No. 1602367.

(38) Tasserit, C.; Koutsoubas, A.; Lairez, D.; Zalczer, G.; Clochard, M.-C. Pink Noise of Ionic Conductance through Single Artificial Nanopores Revisited. *Phys. Rev. Lett.* **2010**, *105*, No. 260602.

(39) Barrett, E. P.; Joyner, L. G.; Halenda, P. P. The Determination of Pore Volume and Area Distributions in Porous Substances. I. Computations from Nitrogen Isotherms. *J. Am. Chem. Soc.* **1951**, *73*, 373–380.

(40) Neimark, A. V.; Lin, Y.; Ravikovitch, P. I.; Thommes, M. Quenched Solid Density Functional Theory and Pore Size Analysis of Microporous Carbons. *Carbon* **2009**, *47*, 1617–1628.

(41) Brunauer, S.; Emmett, P. H.; Teller, E. Adsorption of Gases in Multimolecular Layers. *J. Am. Chem. Soc.* **1938**, *60*, 309–319.

(42) Thommes, M.; Kaneko, K.; Neimark, A. V.; Olivier, J. P.; Rodriguez-Reinoso, F.; Rouquerol, J.; Sing, K. S. W. Physisorption of Gases, with Special Reference to the Evaluation of Surface Area and Pore Size Distribution (IUPAC Technical Report). *Pure Appl. Chem.* **2015**, *87*, 1051–1069.

(43) Knop, A.; Pilato, L. A. *Phenolic Resins - Chemistry, Application and Performance Future Directions*; Springer: Berlin, Heidelberg, 1985.

(44) Moudrakovski, I. L.; Ratcliffe, C. I.; Ripmeester, J. A.; Wang, L.-Q.; Exarhos, G. J.; Baumann, T. F.; Satcher, J. H. Nuclear Magnetic Resonance Studies of Resorcinol–Formaldehyde Aerogels. *J. Phys. Chem. B* **2005**, *109*, 11215–11222.

(45) Mulik, S.; Sotiriou-Leventis, C.; Leventis, N. Time-Efficient Acid-Catalyzed Synthesis of Resorcinol–Formaldehyde Aerogels. *Chem. Mater.* **2007**, *19*, 6138–6144.

(46) Pilato, L. *Phenolic Resins: A Century of Progress*; Pilato, L., Ed.; Springer: Berlin, Heidelberg, 2010.

(47) Henderson, W. A.; Passerini, S. Phase Behavior of Ionic Liquid–LiX Mixtures: Pyrrolidinium Cations and TFSI[−] Anions. *Chem. Mater.* **2004**, *16*, 2881–2885.

(48) Le, M.-L.-P.; Alloin, F.; Strobel, P.; Lepître, J.-C.; Cointeaux, L.; del Valle, C. P. Electrolyte Based on Fluorinated Cyclic Quaternary Ammonium Ionic Liquids. *Ionics* **2012**, *18*, 817–827.

(49) Chen, S.; Liu, Y.; Fu, H.; He, Y.; Li, C.; Huang, W.; Jiang, Z.; Wu, G. Unravelling the Role of the Compressed Gas on Melting Point of Liquid Confined in Nanospace. *J. Phys. Chem. Lett.* **2012**, *3*, 1052–1055.

(50) Hill, A. Microstructural and Molecular Level Characterisation of Plastic Crystal Phases of Pyrrolidinium Trifluoromethanesulfonyl Salts. *Solid State Ionics* **2002**, *154–155*, 119–124.

(51) Kanakubo, M.; Hiejima, Y.; Minami, K.; Aizawa, T.; Nanjo, H. Melting Point Depression of Ionic Liquids Confined in Nanospaces. *Chem. Commun.* **2006**, *17*, 1828–1830.

(52) Buntkowsky, G.; Breitzke, H.; Adamczyk, A.; Roelofs, F.; Emmeler, T.; Gedat, E.; Grünberg, B.; Xu, Y.; Limbach, H.-H.; Shenderovich, I.; Vyalikh, A.; Findenegg, G. Structural and Dynamical Properties of Guest Molecules Confined in Mesoporous Silica Materials Revealed by NMR. *Phys. Chem. Chem. Phys.* **2007**, *9*, 4843–4853.

(53) Jähnert, S.; Vaca Chávez, F.; Schaumann, G. E.; Schreiber, A.; Schönhoff, M.; Findenegg, G. H. Melting and Freezing of Water in Cylindrical Silica Nanopores. *Phys. Chem. Chem. Phys.* **2008**, *10*, 6039–6051.

(54) Giles, J. Electrolytic Conduction in Amorphous Salt Complexed Polyethers. *Solid State Ionics* **1987**, *24*, 155–167.

(55) Choi, U. H.; Ye, Y.; Salas de la Cruz, D.; Liu, W.; Winey, K. I.; Elabd, Y. A.; Runt, J.; Colby, R. H. Dielectric and Viscoelastic Responses of Imidazolium-Based Ionomers with Different Counterions and Side Chain Lengths. *Macromolecules* **2014**, *47*, 777–790.

(56) Lee, D.; Jung, H. Y.; Park, M. J. Solid-State Polymer Electrolytes Based on AB₃-Type Miktoarm Star Copolymers. *ACS Macro Lett.* **2018**, *7*, 1046–1050.

(57) Chu, P. Novel Composite Polymer Electrolyte Comprising Mesoporous Structured SiO₂ and PEO/Li. *Solid State Ionics* **2003**, *156*, 141–153.

(58) Archie, G. E. The Electrical Resistivity Log as an Aid in Determining Some Reservoir Characteristics. *Trans. AIME* **1942**, *146*, 54–62.

(59) Wiedenmann, D.; Keller, L.; Holzer, L.; Stojadinović, J.; Münch, B.; Suarez, L.; Fumey, B.; Hagendorfer, H.; Brönnimann, R.; Modregger, P.; Gorbar, M.; Vogt, U. F.; Züttel, A.; Mantia, F. L.; Wepf, R.; Gróbéty, B. Three-Dimensional Pore Structure and Ion Conductivity of Porous Ceramic Diaphragms. *AIChE J.* **2013**, *59*, 1446–1457.

(60) Ohkubo, T.; Kidena, K.; Ohira, A. Determination of a Micron-Scale Restricted Structure in a Perfluorinated Membrane from Time-Dependent Self-Diffusion Measurements. *Macromolecules* **2008**, *41*, 8688–8693.

(61) Mitra, P. P.; Sen, P. N.; Schwartz, L. M.; Le Doussal, P. Diffusion Propagator as a Probe of the Structure of Porous Media. *Phys. Rev. Lett.* **1992**, *68*, 3555–3558.

(62) Tokuda, H.; Hayamizu, K.; Ishii, K.; Susan, M. A. B. H.; Watanabe, M. Physicochemical Properties and Structures of Room Temperature Ionic Liquids. 1. Variation of Anionic Species. *J. Phys. Chem. B* **2004**, *108*, 16593–16600.

(63) Martinelli, A.; Maréchal, M.; Östlund, Å.; Cambedouzou, J. Insights into the Interplay between Molecular Structure and Diffusional Motion in 1-Alkyl-3-Methylimidazolium Ionic Liquids: A Combined PFG NMR and X-Ray Scattering Study. *Phys. Chem. Chem. Phys.* **2013**, *15*, 5510–5517.

(64) Noda, A.; Hayamizu, K.; Watanabe, M. Pulsed-Gradient Spin-Echo ^1H and ^{19}F NMR Ionic Diffusion Coefficient, Viscosity, and Ionic Conductivity of Non-Chloroaluminate Room-Temperature Ionic Liquids. *J. Phys. Chem. B* **2001**, *105*, 4603–4610.

(65) Tokuda, H.; Hayamizu, K.; Ishii, K.; Susan, M. A. B. H.; Watanabe, M. Physicochemical Properties and Structures of Room Temperature Ionic Liquids. 2. Variation of Alkyl Chain Length in Imidazolium Cation. *J. Phys. Chem. B* **2005**, *109*, 6103–6110.

(66) Zielinski, L. J.; Sen, P. N. Effects of Finite-Width Pulses in the Pulsed-Field Gradient Measurement of the Diffusion Coefficient in Connected Porous Media. *J. Magn. Reson.* **2003**, *165*, 153–161.

(67) Mitra, P. P.; Sen, P. N.; Schwartz, L. M. Short-Time Behavior of the Diffusion Coefficient as a Geometrical Probe of Porous Media. *Phys. Rev. B* **1993**, *47*, 8565–8574.